JALA Academy
Assignment 12

Constructors in Python

Python solution reference • Code examples are organized by the assignment questions.

How to use this document

Save the code examples into a .py file and run with Python 3. Where a program asks for input, enter values when prompted. For file-handling exercises, create the named text file in the same folder first.

1. Default and parameterized constructors

```
class Person:
    def __init__(self, name="Unknown", age=None):
        self.name = name
        self.age = age

    def display(self):
        print("Name:", self.name, "| Age:", self.age)

Person().display()
Person("Anu").display()
Person("Ravi", 22).display()
```

2. Calling parent constructor

```
class Parent:
    def __init__(self, parent_value):
        self.parent_value = parent_value

class Child(Parent):
    def __init__(self, parent_value, child_value):
        super().__init__(parent_value)
        self.child_value = child_value

    def display(self):
        print(self.parent_value, self.child_value)

Child("from parent", "from child").display()
```

3. Public, protected and private naming conventions

```
class AccessDemo:
    def __init__(self):
        self.public_value = "public"
        self._protected_value = "protected by convention"
        self.__private_value = "name-mangled private"

    def private_value(self):
        return self.__private_value

obj = AccessDemo()
print(obj.public_value)
print(obj._protected_value)  # Accessible, but intended for internal/subclass use.
print(obj.private_value())
# obj.__private_value raises AttributeError; Python name-mangles this attribute.
```